Chapter 6

Cover Stories and Hidden Agendas: Early American Space and National Security Policy

Dwayne A. Day

It was language that was perfect in keeping with the popular conception of space at the time, but which was not well-received in the White House. On February 19, 1957, before a collection of spaceflight enthusiasts gathered in San Diego, Major General Bernard Schriever delivered a speech entitled "ICBM—A Step Toward Space Conquest." Schriever was the head of the Western Development Division (WDD), the Air Force development office which was assigned the task of building an operational ICBM as quickly as possible. Schriever's role in developing the ICBM was well known. What was unknown, however, was the fact that WDD was also responsible for developing intelligence satellites. The Air Force planned to launch 92 intelligence satellites over the next five years, culminating in a large signals intelligence satellite in 1962. It was an ambitious plan, but one still lacking money and unknown to the outside world.

Schriever could not speak in San Diego of the classified mission he oversaw. Instead he talked about how WDD's experience in developing the Atlas ICBM could be applied to space. He also spoke of the value of space to the military, and he did not mince words:

> In the long haul our safety as a nation may depend upon our achieving "space superiority." Several decades from now the important battles may not be sea battles or air battles, but space battles, and we should be spending a certain fraction of our national resources to insure that we do not lag in obtaining space supremacy.[1]

The next day, arriving at his headquarters in Los Angeles, Schriever had a telegram from the Office of the Secretary of Defense waiting for him. It was explicit: do not use the word "space" in any more speeches. Remembering back on this event, Schriever's frustration still showed. "I was angry," he recalled over 40 years later while sitting in his office in Washington. But he was an Air Force officer and the Secretary of Defense was his boss. There was never any question about what he would do: "I didn't use the word space anymore in my speeches."[2]

The language in Schriever's speech was not unusual for the times. Wernher von Braun, who had far more visibility as a spokesperson for space than Schriever, had used the term "conquest" often when discussing space, including a series of high-profile articles for *Collliers'* magazine. But where von Braun saw space as new territory to be conquered like the Old West, Schriever saw it as new territory from which to wage battle. He had not been reticent to talk about it in these terms, and despite the muzzling from the Pentagon, the message had already gotten out. On March 4, *Newsweek* ran an article about the American *Vanguard* space program. A small accompanying article included a picture of a stern-looking Schriever and referred to him as "one of the major prophets of space warfare." It quoted the General as stating that "the compelling motive for the development of space technology is ... national defense."[3]

Schriever was only mentioned in the sidebar article, but the main story was unlikely to be any more soothing to the Eisenhower White House, for it was titled "Race Into Space: Can We Win?" The story likely rattled the White House because President Eisenhower and his advisors had explicitly chosen a space policy and a public posture that attempted to downplay the idea the United States was attempting to "race" the Soviets into space. They had chosen a policy which portrayed the fledgling American space program as a small, scientific effort—the development of a rocket and satellite combination referred to as *Vanguard*. Official Pentagon policy was to make no mention of a race and no mention of a military space program.

The only public space program was in actuality a cover, a deception. It shielded what the White House and Pentagon leadership actually cared about, the political vulnerability of its defense

programs. This was not unique to the space program. It was a common strategy for the way that Eisenhower approached many sensitive national security missions—hiding them behind open programs, usually scientific and civilian in appearance, so that they would not be challenged openly by the Soviets. It was to become a common strategy for the later military space program, but it all started with *Vanguard*.

THE KILLIAN REPORT

In September 1954, the Science Advisory Committee of the Office of Defense Mobilization, under orders from President Eisenhower, began a scientific study of the problem of surprise attack.[4] Eisenhower was concerned about recent Soviet developments in various strategic weapons systems and asked the Committee to evaluate American capabilities to respond to them using the latest advances in science and technology. This study was headed by James Killian, then president of the Massachusetts Institute of Technology. The group was known as the Technological Capabilities Panel (TCP); it issued its report, "Meeting the Threat of Surprise Attack," on February 14, 1955. This was often referred to as the "Killian Report" and it greatly impressed Eisenhower. It convinced him that science and technology could be used to defeat an enemy.

During the course of its deliberations, the study's intelligence panel, headed by Polaroid's Edwin "Din" Land, became aware of two advanced proposals for intelligence collection. One was a nuclear powered reconnaissance satellite using a television camera. This was outlined in a report by the Air Force's think-tank, the Rand Corporation, known as *Project Feed Back*. The other idea that Land's panel investigated was a U.S. Air Force development program called BALD EAGLE which was intended to develop a high-altitude reconnaissance aircraft.

While investigating the aircraft program, the panel became aware of another proposal by the Lockheed Skunk Works for its own high-flying strategic reconnaissance aircraft, known as the CL-282. Land brought this proposal to the attention of President Eisenhower in November 1954. Unlike the Air Force program to develop a reconnaissance aircraft, the CL-282 would be configured to carry out strategic reconnaissance prior to hostilities—so-called pre-D-Day reconnaissance.

This was a mission that the Strategic Air Command had previously rejected, but one that those on the TCP thought vital.

Eisenhower approved the CL-282 project and placed it under the charge of the Central Intelligence Agency. The plane eventually became known as the U-2. The program was managed by a newcomer to the CIA, Richard Bissell. The aircraft was never mentioned in the TCP report itself, but was described in a highly classified annex. This annex was for the "Eyes Only" of President Eisenhower, and was probably destroyed by him along with another classified annex on the submarine-launched ballistic missile program.[5]

According to Eisenhower's staff secretary, then-Colonel Andrew Goodpaster, Eisenhower assigned the U-2 mission to the CIA for three reasons. First, he thought it would be less provocative if a civilian pilot, rather than a military one, flew the aircraft into foreign territory. Second, he wanted the product—the reconnaissance photographs—to be evaluated at the national leadership level, as opposed to being evaluated within the military services. Based upon his long years of experience in the military, Eisenhower knew that the military had an incentive to interpret intelligence to its advantage. Finally, he was concerned about not antagonizing the Soviets by pursuing a provocative program in the open. He was concerned that the military would pursue the program in a way that would only escalate tensions between the superpowers.[6] According to Goodpaster, Eisenhower was always distressed to see magazines carrying advertisements of bombers and fighter aircraft. He felt that the military encouraged defense contractors to place such ads and that they projected an image of a society preparing for war with the Soviet Union. This could prompt the Soviets to respond the same way, leading to a spiraling arms race. For Eisenhower, stealth and concealment were the preferable approach, hence the U-2 was given to an organization that could operate in secret.

Those involved in the TCP report realized that overflight of another nation's territory by such an aircraft would constitute a clear violation of international law and could be viewed as a hostile act by the Soviets. In fact, such issues were not abstract, since American aircraft flying missions on the periphery of the Soviet Union were being fired upon on a regular basis and even shot down. The consequences of violating national airspace was clearly a major concern for those planning

aircraft reconnaissance missions. But a satellite, flying much higher, would not necessarily violate international law since no clear definition existed of where "airspace" ended and "space" began. Realizing this, Land and the others on the panel decided to attempt to strongly influence the *establishment* of international law.

The intelligence committee recommended that the United States develop a small artificial earth satellite to establish the right of "freedom of space" for future larger intelligence satellites.[7] Doing so would allow the United States to distinguish between "national airspace" and "international space." By establishing the definition itself, the United States could later use international space to its advantage by flying intelligence satellites over the Soviet Union.[8]

Land, Killian, and others in 1955 considered the reconnaissance satellite to be technologically unrealistic in the near future. They believed that the CL-282 and an Air Force reconnaissance balloon program (later known as *Genetrix*) were more realistic near-term possibilities worthy of the most effort. But they felt that the United States should begin work toward establishing a precedent to enable future satellite reconnaissance missions. The TCP advocated using an ostensibly civilian program to clear a path for a future military program.

THE SCIENTIFIC SATELLITE PROGRAM

In parallel to the deliberations of the TCP, serious proposals for an initial U.S. scientific satellite were emerging. For example, Wernher von Braun and his colleagues at the Army Ballistic Missile Agency in Huntsville, Alabama, teamed up with the Office of Naval Research to propose a satellite called *Orbiter*. Later in the year, the American Rocket Society prepared a detailed survey of possible scientific and other uses of a satellite and proposed it to the National Academy of Sciences' U.S. National Committee for the International Geophysical Year. The IGY was to run from June 1957 to December 1958 and involve international cooperation on the study of the earth.

As it was, 1954 proved to be a very significant year for the generation of ideas concerning scientific and intelligence collection systems. In addition to both the *Feed Back* and CL-282 proposals, the National Academy was now considering a scientific satellite as well. These projects became inextricably politically linked.

The *Feed Back* and the Killian reports were both highly secret, although the *Orbiter* proposal was not. The CL-282 proposal, in particular, was known to only a handful of people.[9] One person who did know of all three projects, as well as of the TCP report, was the Assistant Secretary of Defense for Research and Development, Donald Quarles.

Quarles was also no stranger to proposals for Earth-orbiting satellites. In 1952, Aristid Grosse, a physicist at Temple University, had been asked by President Truman to prepare a report on "the Present Status of the Satellite Problem." The report was not ready before Truman left office and instead the White House forwarded it to Quarles, who as Assistant Secretary of Defense for Research and Development was in charge of virtually all defense-related research projects.

On March 14, 1955 (the same day as the presentation of the TCP report), the U.S. National Committee on the International Geophysical Year of the National Academy of Sciences (NAS) presented a recommendation to Alan Waterman, the Director of the National Science Foundation (NSF). The Committee recommended that a scientific satellite be developed as part of the IGY.[10] After the TCP report came out, Quarles asked Waterman to formally suggest the National Academy's idea to the National Security Council. Four days later Waterman sent a letter to Robert Murphy, Deputy Under Secretary of State, proposing that the United States conduct just such a scientific mission.[11] Those at the National Academy would not have been privy to information about *Feed Back*, CL-282, or the TCP report, so they were unlikely to recognize the strategic objectives behind Quarles' support of their satellite proposal.

Four days later Murphy met with Waterman, Detlev Bronk, President of the NAS, and Lloyd Berkner, one of the Academy's influential members, to discuss the issue. In a letter one month later, Murphy stated that such a proposal would "as a matter of fact, undoubtedly add to the scientific prestige of the United States, and it would have a considerable propaganda value in the cold war."[12] Having gained the support of the Department of State, Waterman then discussed the issue once again with Quarles, who suggested that he consult the Director of Central Intelligence, Allen Dulles, on how best to proceed. Waterman did so and gained Dulles' support for the

program. He also spoke with Percival Brundage, the Director of the Bureau of the Budget, to gain his cooperation when needed. Thus, the scientific satellite proposal now had the support of the Departments of State, Defense, and the Central Intelligence Agency, as well as the Bureau of the Budget. Waterman also agreed to formally propose the full program to an executive session of the National Science Board on May 20.[13] Waterman was lining up bureaucratic support for the project at Quarles' urging.

Whether any of these other people were aware that Quarles' interest in the proposal had been prompted by a classified national security report is unknown. But clearly Quarles was on a mission to enact one of the TCP's recommendations and he worked quickly.

NSC 5520

The speed at which these events took place is startling. Only nine weeks after the TCP report recommended the approval of a scientific satellite program, on May 20, 1955, the National Security Council approved a top-level policy document known as NSC 5520, "U.S. Scientific Satellite Program." The document stated that the United States should develop a small scientific satellite weighing 5 to 10 pounds. It was very clear about why the United States should do this:

> The report of the Technological Capabilities Panel of the President's Science Advisory Committee recommended *that intelligence applications warrant* an immediate program leading to a very small satellite in orbit around the earth, and that re-examination should be made of the principles or practices of international law with regard to "Freedom of Space" from the standpoint of recent advances in weapon technology.[14]

The document continued:

> From a military standpoint, the Joint Chiefs of Staff have stated their belief that intelligence applications strongly warrant the construction of a large surveillance satellite. While a small scientific satellite cannot carry surveillance equipment and therefore will have no direct intelligence potential, it does represent a technological step toward the achievement of the large surveillance satellite, and will be helpful to this end so long as the small scientific satellite program does not impede development of the large surveillance satellite.[15]

NSC 5520 also stated:

> Furthermore, a small scientific satellite will provide a test of the principle of "Freedom of Space." The implications of this principle are being studied within the Executive Branch. However, preliminary studies indicate that there is no obstacle under international law to the launching of such a satellite.
>
> It should be emphasized that a satellite would constitute no active military offensive threat to any country over which it might pass. Although a large satellite might conceivably serve to launch a guided missile at a ground target, it will always be a poor choice for the purpose. A bomb could not be dropped from a satellite on a target below, because anything dropped from a satellite would simply continue alongside in the orbit.

Although the document correctly noted the limited utility of satellites as active military offensive threats, this was not the purpose of the surveillance satellite program. In fact, deploying weapons systems in orbit has never been a significant aspect of American military space policy.

A year later, another White House document stated that the scientific satellite program would have the target date of 1958 "with the understanding that the program developed thereunder will not be allowed to interfere with the ICBM and IRBM programs but will be given sufficient priority by the Department of Defense in relation to other weapons systems to achieve the objectives of NSC 5520."[16] In other words, the scientific satellite would not be allowed to interfere with ballistic missile programs like Atlas and *Jupiter*, but it also would not be allowed to languish without attention while the Department of Defense focused on other programs with more direct military utility. Such inattention would have effectively undercut the entire purpose of NSC 5520. At the same time, the launching of such a satellite was *not* to interfere with the development of the intelligence satellite.

NSC 5520 specifically directed that the scientific satellite program be associated with the International Geophysical Year. "Freedom of space" could be challenged by the Soviets more easily if the U.S. simply launched a satellite at any time. The IGY provided an international context for conducting the program—an excuse. It was a legitimate cover story even better than the civilian auspices of the

existing program. Even if the Soviets had not announced their intention to build a satellite for the IGY, the United States would be able to use the IGY to justify its program if the Soviets complained. Quarles himself specifically noted this reality only a week after the approval of NSC 5520.[17]

The TCP report was perhaps one of the most influential documents of the Cold War. It served as the starting point of a number of major American defense programs in the 1950s. Not only did it recommend the development of what became the U-2, the scientific satellite program, and reconnaissance satellites, but it also recommended the development of what became the Polaris submarine-launched ballistic missile, the *Sosus* underwater sonar array, and the extension of the Distant Early Warning (DEW) Line. The freedom of space recommendation was therefore one among many, and it was in no way particularly special. It was merely one that was unlikely to make progress without specific high-level involvement. Quarles gave it the start that it required.

DEFINING THE HEAVENS

The TCP's extensive list of recommendations required a concerted implementation effort by the various agencies and military services that were affected. It required an extensive oversight effort as well and this task fell to the Operations Coordinating Board of the National Security Council which produced a progress report in June 1955 on the status of the TCP recommendations. This progress report, known as NSC 5522, also included reports from various government departments on their views on the Panel's recommendations. In response to the TCP recommendation concerning freedom of space, NSC 5522 noted that a program had already been started to meet the recommendation. Further it stated:

> State, Treasury, Defense, and Justice Comment: Any unilateral statement by the U.S. concerning the freedom of outer space is unnecessary. It is clear that the jurisdiction of a state over the air space above its territory is limited and that the operation of an artificial satellite in outer space would not be in violation of international law. State and Justice point out that by the convention on international civil aviation of 1944 (to which the U.S. is a party, but the USSR is not) and by customary law every State has exclusive sovereignty "over the air space above its territory." However, air space ends

> with the atmosphere. There has been no recognition that sovereignty extends into airless space beyond the atmosphere.[18]

This statement may have appeared self-evident to those involved, but in effect it begged the question: if space was not sovereign territory (a question for which there was no precedent), how was a nation to determine where the atmosphere—which *was* sovereign territory—ended and space began? The edge of the atmosphere was not a clearly defined boundary and, indeed, could not be defined until satellites began orbiting the Earth to measure the extent of the atmosphere. Still, it made sense for the Departments to advise that the United States not issue a unilateral statement concerning freedom of space, for such a statement would only draw more attention to the subject at a time when the whole point of the scientific satellite program was to establish freedom of space in as inconspicious a manner as possible. For the government lawyers as well as the policy makers, it was better to simply do it and talk about it later.

The CIA also expressed its opinion on the subject in a commentary that proves highly illuminating and remarkably prescient, especially in light of the world reaction to *Sputnik 1* two years later:

> The psychological warfare value of launching the first earth satellite makes its prompt development of great interest to the intelligence community and may make it a crucial event in sustaining the international prestige of the United States.
>
> There is an increasing amount of evidence that the Soviet Union is placing more and more emphasis on the successful launching of the satellite. Press and radio statements since September 1954 have indicated a growing scientific effort directed toward the successful launching of the first satellite. Evidently the Soviet Union has concluded that their satellite program can contribute enough prestige of Cold War value or knowledge of military value to justify the diversion of the necessary skills, scarce material and labor from immediate military production. If the Soviet effort should prove successful before a similar United States effort, there is no doubt but that their propaganda would capitalize on the theme of the scientific and industrial superiority of the communist system.
>
> The successful launching of the first satellite will undoubtedly be an event comparable to the first successful release of nuclear energy in the world's scientific community, and will undoubtedly receive comparable publicity throughout the world. Public opinion in both neutral and allied states will be centered on the satellite's development. For centuries scientists and laymen

> have dreamed of exploring outer space. The first successful penetration of space will probably be the small satellite vehicle recommended by the TCP. The nation that first accomplishes this feat will gain incalculable prestige and recognition throughout the world.
>
> The United States' reputation as the scientific and industrial leader of the world has been of immeasurable value in competing against Soviet aims in both neutral and allied states. Since the war, the reputation of the United States' scientific community has been sharply challenged by Soviet progress and claims. There is little doubt but that the Soviet Union would like to surpass our scientific and industrial reputation in order to further her influence over neutralist states and to shake the confidence of states allied with the United States. If the Soviet Union's scientists, technicians, and industrialists were apparently to surpass the United States and first explore outer space, her propaganda machine would have sensational and convincing evidence of Soviet superiority.
>
> If the United States successfully launches the first satellite, it is most important that this be done with unquestionable peaceful intent. The Soviet Union will undoubtedly attempt to attach hostile motivation to this development in order to cover her own inability to win this race. To minimize the effectiveness of Soviet accusations, the satellite should be launched in an atmosphere of international good will and common scientific interest. For this reason, the CIA strongly concurs in the Department of Defense's suggestion that a civilian agency such as the U.S. National Committee of the IGY supervise its development and that an effort be made to release some of the knowledge to the international scientific community.
>
> The small scientific vehicle is also a necessary step in the development of a larger satellite that could possibly provide early warning information through continuous electronic and photographic surveillance of the USSR. A future satellite that could directly collect intelligence data would be of great interest to the intelligence community.
>
> The Department of Defense has consulted with the [Central Intelligence] Agency, and we are aware of their recommendations, which have our full concurrence and strong support.[19]

The Department of Defense in mid-1955 was not pursuing an active intelligence satellite program. All it had at the time was a series of small Air Force technology development efforts. But the scientific satellite program was underway and the lawyers were now debating the legalities of the program in secret.

If those involved with the legal issues needed their thinking sharpened about issues of international airspace, they could ask for no better lesson than that provided less than a year later by the *Genetrix* reconnaissance balloon program, which had also been endorsed by Killian and Land. Hundreds of balloons were launched beginning in January 1956 so that they would drift over the Soviet Union, their cameras photographing the countryside. Officially, the balloons were part of a scientific program intended to photograph clouds. In reality, they carried sophisticated reconnaissance equipment.

The majority of the balloons were never seen again and American officials knew that they had come down (or been shot down) inside the Soviet Union. This was in reality a rather ironic development; because of the planned future flights of the U-2 at 70,000 feet, the balloons had been ballasted to fly lower than their operational altitude. This was done so that the Soviets would not be provoked into quickly developing a capability to track and shoot down objects at that extreme altitude. The result was that as the balloons cooled at night, they sank below 40,000 feet and well within range of Soviet aircraft, which shot them down in the early morning hours before they rose back to their operational altitude.

On February 7, anticipating the Soviet response, Eisenhower suggested to Secretary of State John Foster Dulles that the operation be suspended and "we should handle it so it would not look as though we had been caught with jam on our fingers." On February 9, the Soviets held a press conference outside Spridonovka Palace. About fifty balloons and instrument containers were placed on display. The balloons, the Soviets said, were part of an espionage project and had been a clear violation of their airspace. It was a major embarrassment for the United States, but the U-2 was still under development and promised far greater capabilities.

Meanwhile, legal issues concerning exactly what "freedom of space" meant occupied U.S. government lawyers. The State Department's Policy Planning Staff was assigned the task of reporting on-going progress on achieving several of the TCP report's recommendations. On October 2, 1956, the Policy Planning Staff reported some preliminary thinking on the issue of freedom of space:

> So far as law is concerned, space beyond the earth is an uncharted region concerning which no firm rules have been established. The law on the subject will necessarily differ with the passage of time and with practical efforts at space navigation. Various theories have been advanced concerning the upper limits of a state's jurisdiction, but no firm conclusions are now possible.
>
> A few tentative observations may be made: (1) A state could scarcely claim territorial sovereignty at altitudes where orbital velocity of an object is practicable (perhaps in the neighborhood of 200 miles); (2) a state would, however, be on strong ground in claiming territorial sovereignty up through the "air space" (perhaps ultimately to be fixed somewhere in the neighborhood of 40 miles); (3) regions of space which are eventually established to be free for navigation without regard to territorial jurisdiction will be open not only to one country or a few, but to all; (4) if, contrary to planning and expectation, a satellite launched from the earth should not be consumed upon reentering the atmosphere, and should fall to the earth and do damage, the question of liability on the part of the launching authority would arise.[20]

Genetrix had provided a good warning of what could happen if an intelligence mission went wrong and if a program's cover was blown. A different lesson was provided in July, when the U-2 made a half dozen sorties over the Soviet Union. The U-2, like *Genetrix*, also had a cover story. It was portrayed as a civilian weather reconnaissance aircraft. But the aircraft went virtually unnoticed outside of the aviation trade press, where some reporters accurately surmised its true purpose. Although the Soviets tracked the plane immediately, they did not protest. Lacking the ability to knock the U-2 out of the sky, they chose not to publicly admit their vulnerability to American reconnaissance technology.

The U-2 demonstrated a number of things to Eisenhower and his advisors. First and foremost, it demonstrated the immense value of overhead reconnaissance systems. The photographs returned by the U-2 were stunning in their quality. The U-2 also demonstrated that both the United States and the Soviet Union had a stake in keeping certain things private. Finally, it demonstrated that a cover story could work. Concealment and obfuscation of intelligence missions had a benefit for superpower relations even when the other side suspected, or actually knew, what was happening.[21] This was the guiding principle of the satellite program adopted under NSC 5520. It was to be a guiding principle for other space programs as well.

SELECTING A SATELLITE PLAN

After the scientific satellite concept had received approval with NSC 5520 in May 1955, it had to be turned into programmatic reality. Not surprisingly, this task fell to Donald Quarles. On June 8, 1955, Secretary of Defense Charles Wilson wrote a memo placing Quarles in charge of starting the scientific satellite program. Quarles then created an Advisory Group on Special Capabilities, chaired by Homer J. Stewart, to select a scientific satellite and a method for launching it. They held hearings and listened to presentations by the Army, Navy and Air Force. The Army's proposal was basically a variation of von Braun's earlier Orbiter concept. The Air Force submitted a proposal from Schriever's Western Development Division, but did not make a concerted effort to win the program. The Navy Research Laboratory proposed using a modified upper atmosphere research rocket. The Advisory Group selected the Navy proposal, which as named *Vanguard*.[22]

Many involved, particularly the Army group, which consisted primarily of Wernher von Braun's "rocket team," were surprised that the Army proposal had lost out. The Army team had extensive experience with ballistic missiles and had begun developing their proposal a year before. When explaining the decision years later, members of the Advisory Group gave a number of reasons, including Navy developments in satellite instrumentation, the supposed lower costs and growth potential of the *Vanguard*, and the desire to avoid using a ballistic missile for the program.[23] The latter point was the most relevant one: NSC 5520 was quite explicit in its statement that the satellite program should not interfere at all with ICBM and IRBM programs. There was no way that the Army proposal could be selected without interfering with Army development of the *Redstone* booster as the basis of an IRBM to be called *Jupiter*—at least without significant additional cost.

The Advisory Group's decision in favor of the Navy program did not settle the issue. The Huntsville group, justifiably, felt that they deserved to win. "There wasn't much confidence, frankly, in the *Vanguard*," said Fred Durant, one of the original members of the *Orbiter* group. The Navy proposal was too new and the team unproven. But according to Durant it was not necessarily arrogance that caused the Army team to be so stunned. The Huntsville group was

more confident because of their longer experience launching rockets and making mistakes.[24] The *Vanguard* team had none of that experience and lacking that, would have to make their own mistakes before achieving success. From the perspective of the Army, that meant that the Soviets were more likely to reach the goal first.

Throughout 1955 and 1956, the Army team proceeded with development of a *Jupiter/Redstone* rocket designed to conduct re-entry tests of ballistic nosecones while the Navy developed the *Vanguard* and the NSF and National Academy of Sciences developed the satellite for flying atop it. But the Army continued to lobby behind the scenes to obtain permission to proceed with their own satellite launching program.

On April 23, 1956, the newly formed Army Ballistic Missile Agency, under the command of Major General John B. Medaris, informed the Office of the Secretary of Defense that a *Jupiter* missile could be fired in an effort to orbit a small satellite as early as January 1957. It proposed that this be considered an alternate to the Navy *Vanguard* program. On May 15, the Secretary of Defense's Special Assistant for Guided Missiles disapproved ABMA's proposal. On May 22, the Assistant Secretary of Defense informed ABMA that no plans or preparations should be initiated for using either *Jupiter* or *Redstone* missiles as satellite launch vehicles.[25]

But all good military officers and bureaucrats know that there is no such thing as a "no" in Washington. ABMA took its plea to Deputy Secretary of State Hoover, who was in charge of the Operations Coordinating Board which was responsible for overseeing the implementation of the NSC 5520 decision. In June 1956, then-Colonel Andrew Goodpaster, who was serving as Eisenhower's staff secretary, reported on an incident that demonstrated both the Huntsville group's lobbying and their perception of the prejudice against them. Goodpaster stated:

> On May 28th Secretary Hoover called me over to mention a report he had received from a former associate in the engineering and development field regarding the earth satellite project. The best estimate is that the present project would not be ready until the end of '57 at the earliest, and probably well into '58. *Redstone* had a project well advanced when the new one was set up. At minimal expense ($2–$5 million) they could have a satellite ready for firing by the end of 1956 or January 1957. The *Redstone* project is one

essentially of German scientists and it is American envy of them that has led to a duplicative project.

I spoke to the President about this to see what would be the best way to act on the matter. He asked me to talk to Secretary Wilson. In the latter's absence, I talked to Secretary [Deputy Secretary of Defense] Robertson today and he said he would go into the matter fully and carefully to try to ascertain the facts. In order to establish the substance of this report, I told him it came through Mr. Hoover (Mr. Hoover had said I might do so if I felt it necessary).[26]

This incident started a new evaluation of the satellite selection process which ultimately rejected the Huntsville group once again. On June 22, 1956, Homer J. Stewart, who was Chairman of the Advisory Group on Special Capabilities, reported the results of two meetings held by the Group in late April to Charles C. Furnas, Assistant Secretary of Defense for Research and Development who had replaced Quarles (who was now Secretary of the Air Force). Stewart reported that although Project *Vanguard* was suffering some minor setbacks and was short of highly capable people, in general the project was on a satisfactory schedule and "one or more scientific satellites can be successfully placed in orbit during the IGY."[27] Stewart also stated:

Redstone re-entry vehicle No. 29, now scheduled for firing in January 1957, apparently will be technically capable of placing a 17 pound payload consisting principally of radio beacons and doppler-equipment in a 200-mile orbit, even with the degradation in performance below the present design figures which might reasonably be expected, but without any appreciable further margin. This capability will depend upon successful accomplishment of several developments, such as the use of a new fuel in the *Redstone* booster, and the spinning cluster of fifteen solid propellant motors. The probability of success of this single flight cannot be reliably predicted now, but it would doubtless be less than 50 per cent.

Stewart explained why the Army proposal should be rejected once again:

In any case, such a single flight would not fulfill the Nation's commitment for the International Geophysical Year because it would have to be made before the beginning of that period. Adequate tracking and observation equipment for the scientific utilization of results would not be available at this time. Moreover, any announcement of such a flight (or worse, any

leakage of information if no prior announcement were made) would seriously compromise the strong moral position internationally which the United States presently holds in the IGY due to its past frank and open acts and announcements as respects *VANGUARD*.

Stewart mentioned that a *Redstone* could be used as a backup later in 1957 if *Vanguard* fell behind schedule and that No. 29 was the only vehicle which could be used without interference in the *Redstone* program. Finally, he concluded, "At the present time, therefore, the Group does not recommend activating a satellite program based on the *Redstone* missile, but will reconsider this question and the possibilities of the ICBM program at its subsequent meetings when the critical items of the *VANGUARD* program are further advanced."[28]

On July 5, 1956, E.V. Murphree, DoD's Special Assistant for Guided Missiles, wrote to Reuben B. Robertson, Jr., the Deputy Secretary of Defense. Murphree stated that he had looked into the matter of using a *Jupiter* re-entry test vehicle for possibly launching a satellite into orbit. He also stated that the January 1957 test could be adapted to this purpose with little effort and no impact on the program and that an attempt could be made as early as September 1956, although this *would* affect the *Jupiter* program. Both dates were before the June 1957 start of the International Geophysical Year.

Murphree further noted that proposals for using the *Jupiter* were not new and that the original *Redstone* satellite and re-entry test vehicle proposals resulted from a common study (the early Orbiter proposal) which argued that the same vehicle could be used for both. He also stated that the first two tests of the *Jupiter* were essentially propulsion system tests and could accomplish much of their goals for that program even if used for satellite launch attempts. He continued: "There is, however, room for serious doubt that two isolated flight attempts would result in achieving a successful satellite, and the dates of such flights would be *prior* to the Geophysical Year for which a satellite capability is specifically required, and prior to the time when tracking instrumentation will be available." (emphasis added)[29]

Murphree said these facts had been taken into consideration at the time that the Office of the Secretary of Defense reviewed the satellite program and decided to assign the mission to the Navy group. He stated: "That decision was based largely on a conviction that the

VANGUARD proposal offered the greater promise for success. The history of increasing demands for funds for this program confirms the conviction that this is not a simple matter. I know of no new evidence available to warrant a change in that decision at this time."

The rest of Murphree's memorandum is extremely interesting and is reprinted below in full:

> While it is true that the *Vanguard* group does not expect to make its first satellite attempt before August 1957, whereas a satellite attempt could be made by the Army Ballistic Missile Agency as early as January 1957, little would be gained by making such an early satellite attempt as an isolated action with no follow-up program. In the case of *Vanguard*, the first flight will be followed up by five additional satellite attempts in the ensuing year. It would be impossible for the ABMA group to make any satellite attempt that has a reasonable chance of success without diversion of the efforts of their top-flight scientific personnel from the main course of the *Jupiter* program, and to some extent, diversion of missiles from the early phase of the re-entry test program. There would also be a problem of additional funding not now provided.
>
> For these reasons, I believe that to attempt a satellite flight with the *Jupiter* re-entry test vehicle without a preliminary program assuring a very strong probability of its success would most surely flirt with failure. Such probability could only be achieved through the application of a considerable scientific effort at ABMA. The obvious interference with the progress of the *Jupiter* program would certainly present a strong argument against such diversion of scientific effort.
>
> On discussing the possible use of the *Jupiter* re-entry test vehicle to launch a satellite with Dr. Furnas, he pointed out certain objections to such a procedure. He felt there would be a serious morale effect on the *Vanguard* group to whom the satellite test has been assigned. Dr. Furnas also pointed out that a satellite effort using the *JUPITER* re-entry test vehicle may have the effect of disrupting our relations with the nonmilitary scientific community and international elements of the IGY group.
>
> I don't know if I have a clear picture of the reasons for your interest in the possibility of using the *Jupiter* re-entry test vehicle for launching the satellite. I think it may be helpful if Dr. Furnas and I discuss this matter with you, and I'm trying to arrange for a date to do this on Monday.[30]

Robertson's special assistant, Charles G. Ellington, forwarded the memos to White House Staff Secretary Andrew Goodpaster. Goodpaster wrote on the bottom of Ellington's cover memo, "Secy.

Robertson feels no change should be made—per Mr. Ellington. Reported to President."[31]

On September 20, 1956, *Jupiter*-C rocket number RS-27 was launched from Cape Canaveral. It flew 3,355 miles, reached an altitude of 682 miles, and achieved a velocity of Mach 18. It carried a deactivated fourth stage that was filled with sand under specific orders from Major General Medaris. Medaris had received specific orders himself. He had been instructed by someone in the Pentagon to ensure that the *Jupiter*'s upper stage was not fueled and did not "accidentally" place a satellite in orbit before the International Geophysical Year. Wernher von Braun had been told to keep quiet, and did so.[32]

Thus, despite the obvious potential to launch a satellite sooner to beat the Soviets into space or simply to maximize the possibility that the United States would place something in orbit, no such effort was made. Despite much high-level discussion of the psychological impact of being first into space, neither the White House or Pentagon officials pursued this feat with great commitment. Indeed, there was no clear desire to simply be first.

HOW TO "BEAT" THE SOVIETS

As one would expect, the scientists involved in creating the program stressed that a scientifically useful program would enhance U.S. prestige. National Science Foundation Director Alan Waterman specifically noted that the schedule was less important than the prestige to be gained from a program that produced a major scientific breakthrough. In other words, being first but not scientifically significant *wasn't* as important for national prestige as accomplishing something scientifically noteworthy.[33] *One could win the race, but lose the scientific competition.*

Rather surprisingly, this was also the conclusion of the National Security Council's Planning Board. NSC 5520 had mentioned the possibility of the Soviets developing a satellite, but took no position on whether the U.S. should attempt to launch before the Soviets. The Defense Department's response in NSC 5522 stated with more urgency that the United States should be first, but must do so carefully. The NSC Planning Board, in November 1956, made an amazing statement:

> The USSR can be expected to attempt to launch its satellite before ours and to attempt to surpass our effort in every way. It is vitally important in terms of the stated prestige and psychological purposes that the United States make every effort to (1) make possible a successful launching as soon as practicable and (2) put on as effective an IGY scientific program as possible. The prestige and psychological set-backs inherent in a possible earlier success and larger satellite by the USSR could at least be partially offset by a more effective and complete scientific program by the United States. Even if the United States achieved the first successful launching and orbit, but the USSR put on a stronger scientific program, the United States could lose its initial advantage.[34]

Thus, one year before Sputnik, the NSC's official position was that the United States could lose prestige *even* if it launched a satellite first but the Soviets developed a better science program. Science, therefore, was a higher priority than schedule.

The issue of the propaganda value of launching a satellite had been mentioned in numerous documents, including Rand reports, intelligence assessments, and NSC 5520 and NSC 5522. In NSC 5522, the CIA equated it with the psychological impact of the atomic bomb. Clearly, many top U.S. policy makers felt that it was a major issue.

But Eisenhower always dismissed these concerns. He just did not believe that it would be that important.[35] His concern was with the political vulnerability of reconnaissance systems, not a race for prestige. He proved to be dreadfully wrong from a domestic political standpoint, but not necessarily from the standpoint of international law.

SPUTNIK CRISIS?

By the fall of 1957, the *Vanguard* satellite program was proceeding roughly on schedule, with first launch anticipated for late 1957 or early 1958, during the International Geophysical Year. No one, not the Huntsville team or the *Vanguard* engineers themselves, expected the first launch to be successful. All rockets blew up the first time they were launched. But *Vanguard* had a year and a half to place a satellite in orbit. A year and a half to fail. The primary criteria was that *Vanguard* achieve orbit before the American intelligence satellite was ready to fly.

The satellite reconnaissance program, however, was underfunded and not making significant progress. Rather surprisingly, this was the fault of Donald Quarles, who had served as Secretary of the Air Force and then risen to become Deputy Secretary of Defense. Although Quarles had immediately appreciated the potential for using a civilian scientific satellite to establish freedom of space for future military satellites, he was skeptical of the reconnaissance satellite then under development by Schriever's Western Development Division. Quarles felt that the program was unlikely to produce anything in the short term and he therefore refused to provide significant funding for development. As many of those involved in the program noted, Quarles created a self-fulfilling prophecy: Quarles thought that the program was not advanced to deserve funding, but the program was not advancing *because* it was underfunded.

Quarles had ordered those in charge of it to not build any actual hardware. Schriever's underlings essentially ignored the order, pushing ahead with a satellite design. But they lacked money. They had asked for $117 million in 1956 and got almost nothing. General Schriever was bewildered. "Now what can I do?" he asked later. "I can't understand it. One of the great problems is surprise attack, and here we're saying we can do a satellite reconnaissance program, and we get $3 million ... ."[36]

Eisenhower's scientific advisors Din Land and James Killian took renewed interest in the reconnaissance satellite by the fall of 1957 and the Science Advisory Committee even sponsored a special briefing for the White House on the subject on September 20.[37] A proposed interim reconnaissance satellite was gaining increased attention from Washington at the same time that the small *Vanguard* satellite was taking shape. The strategy was proceeding, but by October it was overtaken by events.

On October 4, 1957, the Soviet Union launched a 184 pound metallic ball called *Sputnik 1*. It was instantly major news around the world. On October 7, acting on the publicity generated in the wake of *Sputnik 1*, President Eisenhower asked Donald Quarles, by then Deputy Secretary of Defense, to explain why the United States was in the position it was. Goodpaster recorded Quarles' explanation in the minutes of the meeting: "The Science Advisory Committee had felt, however, that it was better to have the Earth satellite proceed separately from military development. One reason was to stress the peaceful character of the effort, and a second was to avoid the inclusion of material, to which foreign scientists might be given access, which is used in our own military rockets." Furthermore, "He [Quarles] went on to add that the Russians have in fact done us a good turn, unintentionally, in establishing the concept of freedom of international space—this seems to be generally accepted as orbital space, in which the missile is making an inoffensive passage."[38]

Two days later, Eisenhower mentioned the subject of getting "beaten" in another staff meeting. "When military people begin to talk about this matter, and to assert that other missiles could have been used to launch a satellite sooner, they tend to make the matter look like a 'race,' which is exactly the wrong impression."[39] As far as Eisenhower and Quarles were concerned, the United States had never been in a race and had certainly not been "beaten." The complex strategy had not gone entirely according to plan, but the end result was the same for them. But Eisenhower also had a blind spot. While he was concerned that taking certain actions like publicly racing the Soviets into space would prompt them into escalating the Cold War, he failed to understand that failing to take these actions could allow others to establish the agenda. He and his advisors were caught in a web of political manipulation that may have been too subtle and complex for their own good.

The true purpose of the early American space program, however, remained shrouded in secrecy for decades. It was only one of many secrets surrounding the program. There were others.

THE CIA AND THE SCIENTIFIC SATELLITE

When Alan Waterman met with Allen Dulles in early May 1955 to discuss NSF sponsorship of a scientific satellite, he also met with Richard Bissell, whom he described as "the one in Central Intelligence who is following this closely."[40] Bissell had reason to follow it closely, since he was then in the middle of managing the newly created U-2 reconnaissance aircraft program. Use of the U-2 hinged on issues of international law and the CIA therefore had good reason to pay attention to any subject involving international airspace.

But even more surprising is the discovery that, as the scientific satellite program continued and ran into significant cost overruns, the CIA actually provided money for it to continue. In April 1957, the Director of the Bureau of the Budget, Percival Brundage, sent a lengthy memo to Eisenhower on the *Vanguard* cost overruns. *Vanguard* was initially supposed to cost 15–20 million dollars. By spring 1957, it was projected to cost ten times that amount. Brundage recounted the funding difficulties of the program and stated, "Apparently, both the Department of Defense and the National Science Foundation are very reluctant to continue to finance this project to completion. But each is quite prepared to have the other do so." He also noted that the National Science Foundation had contributed an extra $5.8 million in funds to the Department of Defense to fund the program and that the CIA had contributed $2.5 million of its own money to the program as well.[41]

Why was the CIA, which had no official stake in the scientific satellite program and no space programs underway or even under study, willing to spend its own money on a civilian scientific space program? The answer is unknown. But it was likely due to Bissell's intervention, since he was the person delegated to follow the program and he was also in charge of the U-2 reconnaissance aircraft.[42] By April 1957, when the CIA provided the money to the scientific satellite program, the U-2 had already made nearly a dozen flights over the Soviet Union, each protested vigorously, but quietly, by the Soviets. Although the CIA did not have a reconnaissance satellite program at this time, it would be given one by Eisenhower less than a year later, a program code-named *Corona*. Not surprisingly, Richard Bissell was placed in charge of that endeavor as well. Bissell was in control of a substantial amount of funding for covert and technical operations at CIA and it is likely that the money to support the scientific satellite program came from these funds or from Director of Central Intelligence Allen Dulles' substantial discretionary account (which may have accounted for up to a sixth of the CIA's budget at the time).

Brundage concluded his memo to Eisenhower by noting that the Air Force had already started its own, much larger, reconnaissance satellite. "Therefore, whether or not the International Geophysical Year satellite project is completed, research in this area will not be dropped." But this missed the point, since the scientific satellite program had less to do with research than with establishing legal precedent.

FURTHER HIDDEN AGENDAS

The Soviets launched *Sputnik 2* on November 5 and it had an even more profound public relations impact than its predecessor. *Sputnik 2* was not only far larger than the *Sputnik 1*, but it also carried a dog, demonstrating a sophistication that belied early administration attempts to downplay the Soviet achievement. The size of the payload was sufficient for the carrying of an atomic bomb, which heightened Americans' fear that the Soviets could now effectively attack the United States. A little over a month later, the U.S. attempt to launch the *Vanguard* satellite ended in embarrassing failure. Clearly, in the realm of space exploration, the Soviets had taken a substantial lead over the United States. From an international legal standpoint, this should have rendered the subject of freedom of space moot, but it did not. Furthermore, the Eisenhower administration continued its practice of using deception to cover its real goals.

In November, newly-appointed Secretary of Defense Neil McElroy proposed centralizing control of the various American space projects then underway, such as *Vanguard* and WS-117L, along with advanced ballistic missile development. McElroy proposed that they be put in a Defense Special Projects Agency (DSPA), which would be responsible for whatever projects the Secretary would assign to it. The idea for the DSPA apparently came from the Science Advisory Council in mid-October, just days after both Sputnik and McElroy's nomination.[43] Eisenhower himself expressed the opinion that a fourth service should be established to handle the "missiles activity."[44] McElroy said that he was weighing the idea of a "Manhattan Project" for antiballistic missiles. The president thought that a separate organization might be a good idea for this problem.[45] In testimony before Congress, Quarles, whom some regarded as an Air Force partisan, stated that long-range, surface-to-surface missiles had been assigned to the Air Force because it possessed the targeting and reconnaissance capabilities to use them, *not* because it was uniquely an Air Force mission.[46] Space could conceivably be treated in the same way.

Although the plan for incorporating ballistic missile development in the new agency was eliminated, the new space agency idea proceeded. The Special Projects Agency would act as a central authority for all U.S. space programs and would essentially contract out missions to the separate services, civilian government agencies, and even universities and private industry. "Above the level of the three military services, having its own budget, it would be able to concentrate on the new and the unknown without involvement in immediate requirements and inter service rivalries." McElroy stated in front of Congress that "the vast weapons systems of the future in our judgment need to be the responsibility of a separate part of the Defense Department."[47] This proposal was placed in a DoD reorganization bill. At this point, it was still assumed that the entire American space program would remain under military control, although at the level of the Secretary of Defense in an office specially created to manage it.

Discussion of the Defense Special Projects Agency continued within the Administration. Its name was changed to the Advanced Research Projects Agency (ARPA) and Eisenhower sent a message to the Congress on January 7, requesting supplemental appropriations for the agency.[48] In early January the newly created President's Science Advisory Committee addressed the issue of ARPA.

On February 7, 1958, James Killian and Din Land, who was also a member of the President's Board of Consultants on Foreign Intelligence Activities, met with Eisenhower and his staff secretary, General Andrew Goodpaster. There they briefed him on the potential of both a recoverable space capsule and supersonic reconnaissance aircraft program, suggesting that in order to speed up the development of a reconnaissance satellite, the U.S. should pursue the recoverable capsule idea as an "interim" solution. Eisenhower accepted this recommendation at that time and the satellite program was soon named "*Corona.*"

An equally important result of this first meeting was the decision to finalize Secretary of Defense McElroy's proposal and create the Advanced Research Projects Agency (ARPA) to house highly technical defense research programs. General Electric executive Roy Johnson was named as its director. Eisenhower decided to give

ARPA control of all military space programs. The military man-in-space program, meteorological programs, and WS-117L would all be turned over to ARPA.

Sputnik also led to the creation of NASA to manage a civilian space program. Although Eisenhower had initially felt that the military could handle the task of space science, he was eventually persuaded that a civilian space agency was needed, in part by Vice President Nixon, who argued that a civilian agency was important for international prestige purposes. A military space agency could not be used as an effective means to "show the flag," particularly if the President was interested in keeping the Cold War from heating up. But a civilian space agency could be used to great effect for psychological purposes. The creation of these new bureaucracies proceeded apace, drawing much attention from the Congress and the public. ARPA was officially in charge of the nation's WS-117L reconnaissance satellite program, whose existence had leaked to the press in the aftermath of Sputnik.

While all this was being done, Eisenhower had also directed that the recoverable satellite program—the program which eventually became known as *Corona*—be handled by a covert team involving the Air Force and the CIA. For many of those who had been involved in it, including those who had proposed it, the recoverable satellite program simply ceased to exist. As far as they knew, WS-117L was the only satellite program that the United States had.

Those in charge of WS-117L eventually split it off into several programs, including the reconnaissance program known as *Sentry* (later *Samos*), an early warning satellite known as Midas, and Discoverer, an engineering and development program which was to include the launching of biomedical payloads such as mice and monkeys. In reality, Discoverer was nothing more than a cover for the *Corona* program, and *Sentry* appears to have been continued at least in part as a sleight of hand—to distract attention away from Discoverer. Canceling it would have been too suspicious and would have raised the ire of the Air Force. Continuing it kept attention elsewhere. While the American public paid attention to NASA and ARPA and *Sentry*, another program materialized into existence unnoticed. It was just like the strategy used earlier with the scientific satellite program.

FREEDOM OF SPACE MARCHES ON

By the first half of 1958 the issue of the militarization of space flared up again due to the expanding space programs. The National Security Council addressed it initially in a classified document in August 1958. Known as NSC 5814/1, the document stated that the United States must "seek urgently a political framework which will place the uses of U.S. reconnaissance satellites in a political and psychological context more favorable to the U.S. intelligence effort." Responding to this, the State Department declared that one of the priorities for the United States was establishing "an acceptable policy framework for the WS-117L program … ."[49] Even within this highly-classified document, *Corona* was nowhere mentioned.

Protection of WS-117L then became the U.S. goal, and the State Department debated it internally and eventually brought it to the United Nations. At the suggestion of the United States, the United Nations created the U.N. Ad Hoc Committee on the Peaceful Uses of Outer Space (COPUOS). The COPUOS became the source of much heated debate over the next several years, as the Soviets refused to participate and instead complained about American nuclear weapons and overseas bases, stating that concessions on these issues were a prerequisite to their participation in COPUOS. Calls by COPUOS for cooperation between the superpowers on space projects were met with derision from the Soviets, who feared that such cooperative efforts would reveal the limitations of their ICBM technology. Much controversy was generated, but no actual policy. All the time that this was being discussed in the United Nations, the newspapers, and in classified State Department meetings, *Corona* continued its development totally unknown to even those at high levels of the government, who only discussed the legal protection of the overt WS-117L program.

The advent of military reconnaissance satellites themselves created their own legal issues. In January 1959, O.G. Villard, Jr., of Stanford University and a member of the National Academy of Sciences and the Space Science Board (SSB) wrote Lloyd V. Berkner, Chairman of the SSB. Villard expressed concern that the U.S. military might attempt to portray its satellite launchings as scientific in nature. This deception could have negative effects on American space science, particularly with regards to international cooperation. Villard stated that it was in the best interests of U.S. scientists that such deception not occur.

On January 28, 1959, Berkner brought the issue to the attention of Killian, who was by now Eisenhower's science advisor. On February 13, Killian in turn mentioned it to NASA Administrator T. Keith Glennan, and Gordon Gray, Eisenhower's Special Assistant for national security affairs. Gray felt that it might require further action by the National Security Council.[50] The results of their discussion of the issue are not known, but this discussion took place over a year after Eisenhower had directed that a small reconnaissance satellite program be peeled away from the bigger reconnaissance program (the one that Villner referred to in his letter) and be conducted covertly. Indeed, the first launch was scheduled to take place later in the month—under the cover of an engineering and scientific program.

The first Discoverer was launched on February 28, 1959. It was in actuality a test of equipment for the *Corona* program. Although the satellite did not reach orbit, the Soviets still protested the flight, decrying its military nature. This prompted Richard Leghorn, the architect of Eisenhower's 1955 "Open Skies" proposal and the president of Itek, which was then managing the manufacture of *Corona* cameras, to draft a proposal for the president titled "Political Action and Satellite Reconnaissance." In it Leghorn stated:

> The problem is not one of technology. It is not a problem of vulnerability to Soviet military measures. The problem is one of the political vulnerability of current reconnaissance satellite programs.
>
> For many years the U.S. has had overflight capabilities (aircraft and balloons) which have been substantially invulnerable to Soviet military countermeasures, but very vulnerable politically. Already the Communists (East Germany) have attacked Discoverer I as an espionage activity, and we can anticipate powerful Soviet political countermeasures to the Discoverer/*Sentry* series.

Leghorn continued:

> What is needed is a program to put reconnaissance satellites "in the white" through early and vigorous political action designed to:
>
> 1. blunt in advance Soviet political countermeasures;
> 2. gain world acceptance for the notion that the surveillance satellites are powerful servants of world peace and security, and are not illegitimate instruments of espionage;
> 3. regain the political initiative of the "open skies" proposal.[51]

Leghorn was fully aware of all American reconnaissance satellite efforts and it is impossible to believe that his proposal was anything other than a continuation of his earlier thinking on overflight. But at this time the United States still had not orbited a covert reconnaissance satellite (and would not for over a year, suffering a string of failures with *Corona*) and was not *overtly* anywhere near flying a reconnaissance satellite. Taking any public position at all on the subject seemed premature at best and foolish at worst. From the White House's point of view it was best to let the political deliberations at the United Nations run their course. Leghorn's proposal went nowhere.

Gary Powers' U-2 reconnaissance aircraft was shot down on May 1, 1960, creating yet another public embarrassment for Eisenhower that in fact revealed a carefully planned and executed strategy to gather intelligence on the Soviet Union—a strategy that like *Genetrix*, *Corona* and *Vanguard* also relied upon a scientific cover story to mask the true mission of the program. Although the downing ruined the upcoming summit, the U-2 had proved an intelligence bonanza for the United States, something which Eisenhower did not wish to emphasize even in his television address to the nation following the incident. [52] He had no desire to reveal America's intelligence coup and no desire to inflame relations with the Soviet Union.

Soviet Premier Khrushchev used the event to maximum propaganda effect, rejecting Eisenhower's renewed proposal for "Open Skies." Khrushchev declared, "As long as arms exist our skies will remain closed and we will shoot down everything that is there without our consent." France's President de Gaulle then asked whether this would include satellites, noting that Soviet satellites had already carried cameras into space. Khrushchev then replied, "As for sputniks, the U.S. has put up one that is photographing our country. We did not protest; let them take as many pictures as they want."[53]

The State Department continued to discuss the subject in secret. By mid-1960, the Bureau of European Affairs at the State Department had even drafted a policy paper concerning the planned upcoming launch of the *Samos* reconnaissance satellite. One option proposed was an "open approach" which advocated the sharing of all reconnaissance photographs taken by *Samos*. This approach was considered "more likely to facilitate wide acceptance of photographic satellites than the 'closed' approach."[54] While the State Department was debating the merits of sharing reconnaissance satellite photographs with the world community (something that the United States had not done even with U-2 photographs), *Corona's* engineers were preparing for another launch. The memo was written August 12. Six days later *Corona* returned its first images of the Soviet Union. There was no talk in the White House of sharing the photos with anyone.

The State Department continued to debate the merits of the American space program, but gradually even *Samos* itself was enveloped within the dark cloak of secrecy that surrounded *Corona*. State Department discussions in general, and public statements in particular, ceased. "Freedom of space" had essentially been achieved. Further discussion was largely irrelevant. And *Corona* began to chalk up success after success, as the press focused on *Samos* and the public paid attention to NASA and its daring Mercury astronauts.

CONCLUSION

Any close study of the Eisenhower presidency reveals a president who frequently rejected overt and provocative policies in favor of covert actions intended to achieve the same result but without heightening Cold War tensions. It also reveals a President who saw little separation between national security and science and who was fully willing to use allegedly "civilian" science programs to cover military operations. Reconnaissance is perhaps the premier example of this. First with the U-2, then with *Corona*, Eisenhower chose to change direction and place the programs under different management primarily to avoid public scrutiny. He allowed the existing programs to continue before they slowly disappeared. He also used scientific cover stories extensively to cover the reconnaissance programs. Few expected these cover stories to last forever. They were, in the language of the intelligence community, "melting assets." But they worked for a time. Rather surprisingly, they did not appear to have any negative effects for the civilian scientific community.

There is no evidence among Eisenhower's scientist advisors of any real debate over this practice of using science as cover for intelligence programs. They seem to have not recognized any line between the scientific establishment and the national security establishment. The idea of the scientific establishment deliberately separating itself from the military establishment did not come until later. And perhaps most ironically of all, this deliberate blurring of the lines between scientific and military establishments that Eisenhower did so much to create was one of the things he warned of in his farewell address.

1. Major General Bernard A. Schriever, Commander, Western Development Division, "ICBM—A Step Toward Space Conquest," February 19, 1957.
2. General Bernard A. Schriever, interview by Dwayne A. Day, July 17, 1997.
3. "For Defense," *Newsweek*, March 4, 1957, p. 66.
4. J.R. Killian, Jr., to General Curtis E. LeMay, September 2, 1954, Papers of Curtis LeMay, Box 205, Folder B-39356, Manuscript Division, Library of Congress.
5. Information on the classified annexes comes from an interview by Donald E. Welzenbach with James Killian and is referenced in: Donald E. Welzenbach, "Science and Technology: Origins of a Directorate," *Studies in Intelligence*, Vol. 30, Summer 1986, RG 263, National Archives and Records Administration (hereafter referred to as NARA). Although the intelligence section of the TCP report remains classified and awaits review, the index has been declassified. It includes the word "satellites," but apparently in the context of satellite countries of the USSR "The Report to the President by the Technological Capabilities Panel of the Science Advisory Committee, February 14, 1955, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Alphabetical Subseries, Box 16, "Killian Report—Technological Capabilities Panel (2)", Dwight D. Eisenhower Library (hereafter referred to as DDE).
6. General Andrew Goodpaster, interview by Dwayne A. Day, March 19, 1996. Goodpaster went to the White House in October 1954 as a Colonel and was promoted to Brigadier General while there. He eventually rose to the rank of General and assumed command of Supreme Headquarters Allied Powers Europe (SHAPE) in 1969.
7. Although the majority of the contents of the intelligence section of the TCP report remain classified, crucial portions were printed in other documents which have now been declassified. For instance, a cover letter accompanying the report upon its delivery to the Policy Planning Staff at the Department of State declares that one of the intelligence panel's conclusions was the need for the "re-examination of the principles of freedom of space, particularly in connection with the possibility of launching an artificial satellite into an orbit about the earth, in anticipation of use of larger satellites for intelligence purposes." See "Report to the President on the Threat of Surprise Attack," Policy Planning Staff, Department of State, March 14, 1955, General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 87, "NSC 5522 Memoranda", RG 59, NARA.
8. Another document, dated only two weeks later, called for the Department of State to discuss a TCP recommendation on freedom of space. It stated: "The present possibility of launching a small artificial satellite into an orbit about the earth presents an early opportunity to establish

a precedent for distinguishing between 'national air' and 'international space,' a distinction which could be to our advantage at some future date when we might employ larger satellites for intelligence purposes." Robert R. Bowie, "Memorandum for Mr. Phleger," Policy Planning Staff, Department of State, March 28, 1955, Department of State Central Files, 711.5/3-2855.

9. In fact, Schriever, who was responsible for one of the most advanced weapons development projects in the United States military, did not learn of it until later, when he overheard someone discussing it on the phone and attempting to conceal its identity.
10. Joseph Kaplan, Chairman, United States National Committee, International Geophysical Year 1957–58, National Academy of Sciences, to Dr. A.T. Waterman, Director, National Science Foundation, March 14, 1955.
11. Alan T. Waterman, Director, Memorandum for Mr. Robert Murphy, Deputy Under Secretary of State, March 18, 1955.
12. Robert Murphy, "Memorandum for Dr. Alan T. Waterman, Director, National Science Foundation," April 27, 1955.
13. Alan T. Waterman, Director, to Donald A. Quarles, Assistant Secretary of Defense (Research and Development), May 13, 1955.
14. The newly released part of the document is in italics. NSC 5520, May 20, 1955, General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 112, "NSC 5520", RG 59, NARA.
15. Ibid. This portion of the document remained classified until 1995.
16. "James S. Lay, Jr., Executive Secretary, National Security Council, Memorandum for the National Security Council, "U.S. Scientific Satellite Program," November 9, 1956, General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 86, "NSC 5520—US Scientific Satellite Program (Memoranda)", RG 59, NARA.
17. Memorandum, "Discussion at the 250th Meeting of the National Security Council, Thursday, May 26, 1955," May 27, 1955, Files of Dwight D. Eisenhower as President, Ann Whitman File, NSC Series, Box 6, "250th Meeting of NSC, May 26, 1955," DDE.
18. National Security Council NSC 5522, June 8, 1955, Comments on the Report to the President by the Technological Capabilities Panel, p. S-5, White House Office, Office of the Special Assistant for National Security Affairs: Records, 1952–61, NSC Policy Papers, Box 16, Folder NSC 5522 Technological Capabilities Panel, DDE.
19. National Security Council NSC 5522 June 8, 1955 Comments on the Report to the President by the Technological Capabilities Panel, p. A-55–6.
20. Robert R. Bowie, Policy Planning Staff, Department of State, "Recommendations in the Report to the President by the Technological Capabilities Panel of the Science Advisory Committee, ODM (Killian Committee): Item 2—NSC Agenda 10/4/56," General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 87, "NSC 5522 Memoranda," RG 59, NARA.
21. These "gentlemen's agreements" were a tenet of the Cold War. They were certainly common in the field of espionage, where the superpowers frequently chose to keep quiet when they caught the other side spying. They were also part of a larger unwritten code of cooperation between the two nations, such as the tacit, unspoken agreement not to seek the assassination of the other side's leaders.
22. Constance McLaughlin Green and Milton Lomask, *Vanguard: A History* (Washington, DC: Smithsonian Institution Press, 1971), pp. 34–56.
23. Ibid., p. vi. Green and Lomask added: "To these observations, I can add from my own experience that inter-service rivalry exerted strong influence; also, that any conclusion drawn would be incomplete without taking into account the antagonism still existing toward von Braun and his co-workers because of their service on the German side of World War Two."
24. Fred Durant, interview by Dwayne A. Day, July 27, 1997.
25. *Redstone Arsenal Complex Chronology, The ABMA/AOMC Era, 1956–62*, Redstone Arsenal webpage.
26. Colonel A.J. Goodpaster, "Memorandum for Record," June 7, 1956, White House Office, Office of the Staff Secretary: Records, 1952–1961, Box 6 "Missiles and Satellites," DDE.
27. Homer J. Stewart, Chairman, Advisory Group on Special Capabilities, Memorandum for the Assistant Secretary of Defense (R&D), "VANGUARD and REDSTONE," June 22, 1956, White House Office, Office of the Staff Secretary: Records, 1952–1961, Box 6, "Missiles and Satellites," DDE.
28. Stewart's memorandum was stamped "SECRET," but there is some doubt as to whether it was actually written in *May* 1956 instead of June. It is rare for a report of a meeting to be written two months after the meeting. Furthermore, the memo also mentions the Group's *upcoming* meeting on June 19 and 20 concerning the propulsion systems for Vanguard and invites contractor representatives to attend this meeting, which would already have happened by the time the memo was written. The June 22 date may be a typo.
29. E.V. Murphree, Special Assistant for Guided Missiles, Memorandum for Deputy Secretary of Defense, "Use of the JUPITER Re-entry Test Vehicle as a Satellite," July 5, 1956, White House Office, Office of the Staff Secretary: Records, 1952–1961, Box 6, "Missiles and Satellites," DDE.
30. In another brief, one-page memorandum from C.C. Furnas to the Deputy Secretary of Defense, dated July 10, 1956 and stamped "Secret," Furnas mentioned the meeting that he and Murphree had with Robertson on July 9. Furnas used this memorandum as a cover letter to forward the previous report to him by Homer Stewart's Advisory Group on Special Capabilities. He concluded by saying "I trust that this will serve your purpose in reporting your evaluation of the suggestion that a Redstone vehicle will be used." C.C. Furnas, Assistant Secretary of Defense for Research and Development, Memorandum for Deputy Secretary of Defense, July 10, 1956, White House Office, Office of the Staff Secretary: Records, 1952–1961, Box 6, "Missiles and Satellites," DDE.
31. William Ewald, *Eisenhower the President: Crucial Days 1951–1960* (Englewood Cliffs, NJ: Prentice-Hall, 1981), p. 284.
32. Ernst Stuhlinger, letter to Dwayne A. Day, August 23, 1997.
33. Alan T. Waterman, National Science Foundation, Memorandum to Mr. Percival Brundage, Bureau of the Budget, "Funding of Earth Satellite Program, International Geophysical Year," April 7, 1956, General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 86, "NSC 5520—U.S. Scientific Satellite Program (Memoranda)," RG 59, NARA.
34. James S. Lay, Jr., Executive Secretary, National Security Council, Memorandum for the National Security Council, "U.S. Scientific Satellite Program," November 9, 1956, with attached: "Draft Report on NSC 5520, U.S. Scientific Satellite Program Background," General Records of the Department of State: Records Relating to State Department Participation in the Operations Coordinating Board and the National Security Council, 1947–1963, Box 86, "NSC 5520—US Scientific Satellite Program (Memoranda)," RG 59, NARA.
35. Goodpaster interview.
36. General Bernard Schriever interview by ArcWelder Films.
37. David Z. Beckler, Executive Officer, Science Advisory Committee, Memorandum for General Goodpaster, "Special Briefing," September 19, 1957, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher

H. Russell, 1952–61, Subject Series, Alphabetical Subseries, Box 23, "Science Advisory Committee (2) Sept.-Oct. 1957, " DDE.

38. Brigadier General Andrew Goodpaster, Memorandum of Conference with the President, October 7, 1957, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 6, "Missiles and Satellites." DDE.
39. Brigadier General Andrew Goodpaster, Memorandum of Conference with the President, (following McElroy swearing in) October 9, 1957, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 6, "Missiles and Satellites," DDE.
40. Waterman to Quarles, May 13, 1955.
41. Percival Brundage, Director, Bureau of the Budget, Memorandum for the President, "Project VANGUARD," April 30, 1957, White House Office, Office of the Staff Secretary: Records, 1952–1961, Box 6 "Missiles and Satellites," DDE.
42. Former CIA Deputy Director of Science & Technology Albert "Bud" Wheelon has speculated that the money probably came from Allen Dulles' substantial discretionary DCI budget.
43. Goodpaster interview.
44. Eisenhower's comments on this subject appear in numerous documents. For instance, in October 1957 Goodpaster reported "The President went on to say he sometimes wondered whether there should not be a fourth service established to handle the whole missiles activity." Brigadier General A.J. Goodpaster, "Memorandum of Conference with the President, October 11, 1957, 8:30 AM," October 11, 1957, Ann Whitman File, DDE Diary Series, Box 67, "Oct. 57 Staff Notes (2)," DDE. In January 1958 Goodpaster reported "In the course of the discussion the President indicated strongly that he thinks future missiles should be brought into a central organization." Brigadier General A.J. Goodpaster, Memorandum of Conference with the President, January 21, 1958," January 22, 1958, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 6, "Missiles and Satellites, Vol. II (1) [January-February 1958]" DDE. In February 1958, Goodpaster reported "The President said that he has come to regret deeply that the missile program was not set up in OSD rather than in any of the services." Brigadier General A.J. Goodpaster, "Memorandum of Conference with the President, February 4, 1958 (following Legislative Leaders meeting)," February 6, 1958, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 6, "Missiles and Satellites, Vol. II (1) [January-February 1958]," DDE.
45. Brigadier General A.J. Goodpaster, Memorandum of Conference With the President, October 11, 1957, Ann Whitman File, DDE Diary Series, Box 27, "Oct. 57 Staff Notes (2)," DDE. In February, another memo states "The President said that he has come to regret deeply that the missile program was not set up in OSD rather than in any of the services. Personal feelings are now so intense that changes are extremely difficult." Brigadier General A.J. Goodpaster, Memorandum of Conference With the President, February 4, 1958 (Following Legislative Leaders meeting)," February 6, 1958, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 8, "Missiles and Satellites, A National Integrated Missile and Space Vehicle Development Program, December 10, 1957," DDE.
46. Robert Frank Futrell, *Ideas, Concepts, Doctrine, Vol. 1*, (Washington, DC: U.S. Government Printing Office, 1989), p. 589. The comments on Quarles' partisanship come from the interview with Goodpaster.

47. *Organization and Management of Missile Programs, Hearings before a Subcommittee of the Committee on Government Operations*, U.S. House of Representatives, 86th Cong., 2d Sess. (Washington, D.C.: U.S. Government Printing Office, 1959), p. 133.
48. Ibid.
49. McDougall, *The Heavens and the Earth*, pp. 182–83.
50. Gordon Gray, Special Assistant to the President, to Brigadier General Andrew J. Goodpaster, February 16, 1959, with attached: James R. Killian, Jr., to Gordon Gray, Special assistant to the President, February 13, 1959; James R. Killian, Jr., to Dr. T. Keith Glennan, Administrator, National Aeronautics and Space Administration, February 13, 1959; Lloyd V. Berkner, Chairman, Space Science Board, National Academy of Sciences, to Dr. James R. Killian, Jr., Special Assistant to the President for Science and Technology, January 28, 1959; O.G. Villard, Jr. Space Science Board, to Dr. L.V. Berkner, President, Associated Universities, Inc., January 22, 1959, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 15, "Space [January-June 1959]," DDE.
51. Richard S. Leghorn, Political Action and Satellite Reconnaissance [Draft], April 24, 1959, Office of the Staff Secretary: Records of Paul T. Carroll, Andrew J. Goodpaster, L. Arthur Minnich and Christopher H. Russell, 1952–61, Subject Series, Department of Defense Subseries, Box 15, "Space [January-June 1959]," DDE.
52. The U-2 also raised once again the issue of where airspace ended and space began. At the Eleventh International Astronautical Federation Congress in Stockholm, Sweden, Spencer M. Beresford presented a paper which connected the U-2 and violations of international airspace with the possibility of future flights by military MIDAS and SAMOS flights. A State Department official obtained a copy of Beresford's paper before his presentation and notified the U.S. Information Service in Stockholm that the paper raised a number of "highly sensitive" topics which the United States should not comment. W.E. Gathright, to USIS-Stockholm, "TOUSI II, Joint State USIA Message," August 12, 1960, with attached: Remarks of Spencer M. Beresford, United States of America, at the Eleventh Annual Congress of the International Astronautical Federation, Stockholm, Sweeden, August 16, 1960, General Records of the Department of State, Bureau of European Affairs, Office of Soviet Union Affairs, Subject Files, 1957–1963, Box 6, "12 Satellite and Missile Programs," RG 59, NARA.
53. Khrushchev was referring to the U.S. Tiros weather satellite launched in August.
54. Foy D. Kohler, Bureau of European Affairs, to Phillip J. Farley, "SAMOS," July 18, 1960, General Records of the Department of State, Bureau of European Affairs, Office of Soviet Union Affairs, Subject Files 1957–1963, Box 6, "12 Satellite and Missile Programs," RG 59, NARA.